

1.2 Numerical illustration

At $\beta = \delta = 4/5$, the first-order condition becomes $12 - 20\hat{c} = 5\hat{c}\sqrt{9 - 10\hat{c}}$. Squaring yields $(5\hat{c} - 4)(50\hat{c}^2 + 75\hat{c} - 36) = 0$; checking the unsquared equation and feasibility leaves the positive root of the quadratic. Its discriminant is $75^2 + 4(50)(36) = 225 \cdot 57$, which explains the appearance of $\sqrt{57}$:

$$\hat{c}^* = \frac{3(\sqrt{57} - 5)}{20} = .382475, \quad p_1^* = \frac{11 + \sqrt{57}}{40} = .463746, \quad p_2^* = \frac{1 + 3\sqrt{57}}{40} = .591238.$$

The optimal rescue menu completes 25.9581% of requests, compared with 25.0000% under the best flat price.

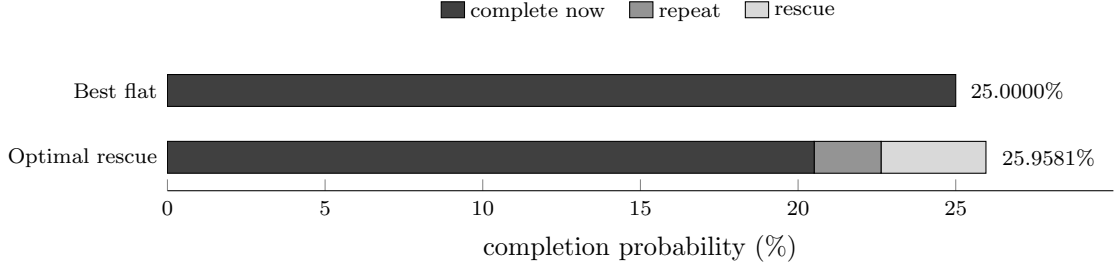


Figure 2: Exact-one-driver completion at $\beta = \delta = .8$. The 0.9581 pp gain combines less completion now with repeat and rescue completion later.

β	δ	$\hat{c}^*$	p_1^*	p_2^*	M_1^*
.6	.5	.460069	.467733	.530034	.253038
.6	.8	.478613	.488210	.539306	.251089
.8	.5	.370421	.420148	.585210	.272458
.8	.8	.382475	.463746	.591238	.259581
.9	.5	.329857	.405843	.614928	.286282
.9	.8	.314018	.453693	.607009	.266932

Each row re-solves the WPBE and optimal menu. More patient riders support a larger rescue region; less patient incumbents make waiting less attractive and increase the completion value of escalation.